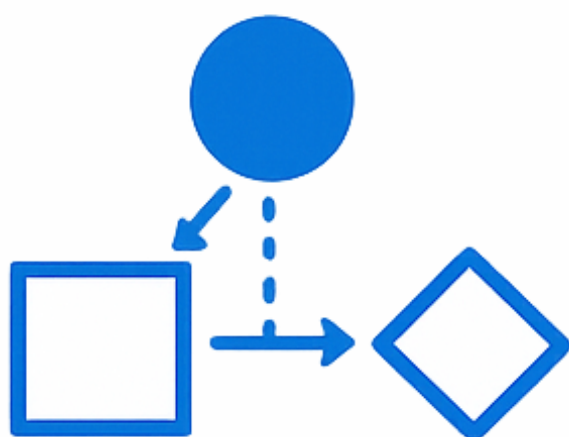




Assignment 2

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Qualitative Methods and Data Quality

Part I: Say-Do Gap Study and Part II: Radiology Data Quality Analysis

Date: March 2, 2026

Course: NDAB20010U Health Data and Interoperability

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AI Declaration

I declare that AI has only been used for grammatical correction.

AI has **not** been used to generate content, reasoning, or structure.

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Abbreviation Key

Table 1: Internal abbreviations and identifiers used in this report

Abbreviation	Meaning in this report
Axx	Activity identifier in the interview-based model in Part I (for example, A01–A09).
Exx	Event identifier in the observed Part I event log (for example, E03, E10).
Dxx	Delta identifier in the Part I comparison sheet (for example, D01–D06).
Mx	Method identifier in the Part II data quality toolkit (for example, M2, M10).
RQ1	Research Question 1: workflow efficiency and bottlenecks.
RQ2	Research Question 2: timeliness of critical finding communication.
RQ3	Research Question 3: AI-based triage system development.
DQ	Data quality.
TAT	Turnaround time between selected workflow milestones.

1 Part I: A Real-World Study of the Say-Do Gap in Modelling

This part analyses one weekday morning routine across *work-as-imagined*, *work-as-done*, and *work-as-recorded*, with explicit links between claims and evidence.

1.1 Deliverable 1: Description of the Routine Based on the Interview

Task Requirement (Deliverable 1) *Describe the routine from the interview and identify activities, their order, and participants.*

Scope and evidence This Part I study is a single-case self-study with one participant (the author) and one routine (weekday morning preparation before leaving home). The interview was conducted on February 18, 2026 as a semi-structured self-interview.

Following Chapter 27, this step is treated as *manual knowledge acquisition*: it captures *knowledge-as-stated* (what the participant reports) and produces an initial *work-as-imagined* model (how the routine is expected to unfold) (Guide to Health Informatics 2026; Zajac 2026). Internal identifiers used in the report are summarised in Table 1.

The routine boundary is anchored in transcript evidence (Appendix A, Table 3, lines L21–L22):

- Start: the alarm sounds.
- End: the apartment door closes when leaving.

The baseline sequence is grounded in lines L02–L12: wake-up, hygiene, dressing/packing, breakfast, final checks, and departure. Variation points (snooze, breakfast substitution, key search, occasional roommate coordination) are grounded in lines L14–L18.

Participants, order, and modelling assumptions Table 4 presents the interview-based activity model, and Figure 1 shows the corresponding interview-based DCR graph. In this context, interview-based DCR means a process model built from interview evidence rather than direct observation.

Assumptions used in Deliverable 1 are:

- The model represents a weekday baseline, not weekend behaviour.
- The participant is the only consistent human actor in the routine.
- Expected times are recall-based estimates and therefore approximate.

Together, these assumptions keep scope explicit and make later comparison with observed evidence in Deliverables 2 and 3 auditable.

1.2 Deliverable 2: Event Log Representing the Observed Routine

Task Requirement (Deliverable 2) *Create an event log of the observed routine as a CSV file with timestamps, event labels, and relevant context notes.*

The observation was conducted on February 20, 2026 (06:45–07:30), two days after the interview. This delay was intentional to reduce recall carry-over between elicitation and observation.

In methodological terms, the observation captures *work-as-done* (what was actually performed in context). The resulting CSV captures *work-as-recorded* (what is represented in structured trace data) (Guide to Health Informatics 2026; Zajac 2026). To keep data collection transparent, the observation followed a short guide (Appendix A, Table 2).

The submitted event log includes the fields `case_id`, `event_id`, `timestamp`, `event_label`, `actor`, `tool`, `context_note`, and `related_activity_id`. The full machine-readable file is submitted separately, while Appendix A, Table 5 shows the key excerpt used in the report.

Compared with the interview baseline, the observation reveals three anchor differences that are used in Deliverable 3:

- E03: extra coordination event,
- E09: order shift relative to the interview sequence,
- E10: explicit workaround behaviour.

These observations provide the empirical basis for the delta analysis between stated and observed practice.

1.3 Deliverable 3: Delta Sheet Summarising Differences Between Stated and Observed Practice

Task Requirement (Deliverable 3) *Compare Deliverables 1 and 2 and produce a delta sheet with missing events, extra events, different order, timing distortions, workarounds/coordination steps, and relabeling/ambiguity.*

The comparison maps interview activities (Axx identifiers) to observed events (Exx identifiers). A delta is registered when at least one of the following differs: event presence, order, timing, workaround behaviour, or activity interpretation. This follows the Chapter 27 logic of comparing observed traces with a prior process description to expose model limits and recurring exceptions (Guide to Health Informatics 2026).

Appendix A, Table 6 contains the complete delta sheet. The six required categories are all represented:

- Missing event: no distinct seated breakfast-and-schedule-review step.

- Extra event: roommate coordination appears only in observed data.
- Different order: bag packing occurs after breakfast preparation starts.
- Timing distortion: hygiene begins later than the interview baseline.
- Workaround/coordination step: key/lock issue triggers ad hoc messaging.
- Relabeling/ambiguity: *breakfast* changes from sit-down meal to banana plus coffee-to-go.

These deltas operationalise the say-do gap, meaning the difference between described routine and performed routine. They also prepare the reflection in Deliverable 4 by showing where model assumptions break under real execution conditions.

1.4 Deliverable 4: Reflections on the Modelling Process

Task Requirement (Deliverable 4) *Reflect on six modelling questions on missed data, activity definition, event-log limits, automation potential, observed variations, and the role of qualitative discovery.*

1. **Question 1.** *What kinds of data were missed or misrepresented in the interview-based model and why?*

Answer

Claim. The interview-based model captured intent, but underrepresented situated coordination and recovery behaviour.

Evidence. Observation added micro-coordination (E03), workaround behaviour (E10), and omission of the planned seated breakfast-and-schedule step.

Implication. Interview-only modelling overstates routine linearity and should be treated as a starting model, not a full process description.

2. **Question 2.** *What counts as an activity in each model, and how did this differ?*

Answer

Claim. The two models use different granularity.

Evidence. Deliverable 1 uses intention-level activities (Axx), while Deliverable 2 logs timestamped observable events (Exx).

Implication. One interview activity can map to multiple observed events, so apparent stability at intention level can hide operational variation.

3. **Question 3.** *How did the event log constrain what could be modelled?*

Answer

Claim. The event log supports sequence and timing analysis well, but limits interpretation of motives and local judgement.

Evidence. The log records *what* happened and *when*, while causal context is only partially recoverable from notes.

Implication. Process modelling from logs alone can identify deviations, but cannot fully explain why deviations occurred.

4. **Question 4.** *Do you think that the event log could be captured automatically by a system? Why or why not?*

Answer

Claim. Automatic capture is feasible for digital traces but incomplete for social and tacit work.

Evidence. Timestamped app events can be captured reliably, but ad hoc coordination and workaround decisions are often off-system.

Implication. Automated logging should be combined with targeted qualitative follow-up when model validity depends on context.

5. **Question 5.** *How did the observation help uncover variations or exceptions that were not mentioned in the interview?*

Answer

Claim. Observation exposed exceptions that were absent from the stated routine.

Evidence. The observed trace showed order changes (E07 before E09), a missing distinct A06 step, and workaround handling (E10).

Implication. Conformance checking should be paired with exception analysis, because recurring local deviations can be practically rational.

6. **Question 6.** *Why is it relevant to consider qualitative methods of discovery (e.g., observations) when building process models?*

Answer

Claim. Qualitative methods improve model validity by grounding formal traces in real practice.

Evidence. The combined use of interview, observation, and event log produced a more coherent account than any single source alone.

Implication. Robust process models require triangulation across stated, observed, and recorded evidence (Guide to Health Informatics 2026; Zajac 2026).

2 Part II: Radiology Case Study — Data Quality, Conformance, and the Limits of Logs

This part synthesises notebook evidence (Deliverable 5) and the accompanying Part II reflection section (Deliverable 6).

2.1 Deliverable 5: Notebook Submission

Task Requirement (Deliverable 5) *Submit the completed notebook `assignment.ipynb` with the full Part II workflow and method execution.*

The completed notebook is submitted as a separate artefact (`assignment.ipynb`) in the required folder.

It includes all Part II steps (Parts 0–7): initial exploration, selection and execution of data quality (DQ) methods, custom DQ dimension design, per-facet synthesis, and invisible-work analysis.

Method identifiers (Mx) and research-question identifiers (RQ1–RQ3) follow the abbreviation key in Table 1.

2.2 Deliverable 6: Part II Reflection Questions

1. **Initial exploration. What kind of data is logged in the file? Assuming data quality is good, how can the data be used to answer the RQs?**

Answer

The file is a process event log from a radiology workflow. Each row represents an event linked to a case (`case_id`) and includes an event timestamp, activity label, actor, role, and optional clinical/technical context fields such as modality and report metadata. From the Part 1 exploration (Appendix B, Table 7), the dataset contains 150 events across 17 cases, covering 12 unique logged activities (Appendix B.5, Table 12).

Assuming good data quality, this type of log can support all three RQs directly:

- **RQ1 (workflow efficiency/bottlenecks):** timestamps and ordered activities can be used for turnaround-time calculations, handoff delays, and bottleneck localization.
- **RQ2 (critical finding communication):** the presence and timing of `Critical communication` events can be checked against safety-relevant conformance expectations.
- **RQ3 (AI triage development):** case- and event-level features (timing, urgency proxies, role patterns, workflow pathways) can be extracted as candidate training signals.

So the basic data structure is appropriate for the assignment goals, but only if completeness, representativity, timeliness, and provenance are sufficiently trustworthy.

2. **Method selection justification. Include your completed method selection table from Part 3. Additionally, answer why you did not choose the other methods, what trade-offs were considered, and how RQs influenced selection.**

Answer

I selected five methods: **M2, M6, M8, M10, and M11**. The completed selection table from Part 3 is included in Appendix B, Table 8. The main logic was to cover all three RQs with complementary methods and minimal redundancy:

- **M2** gives a baseline quality gate (what fields can be trusted at all).
- **M6** evaluates representativity of case context (urgency/shift coverage), important for generalisability.
- **M8** directly targets safety-critical communication behavior for RQ2.
- **M10** links operational performance metrics to timestamp credibility, which is central for RQ1.
- **M11** adds human/workload structure that matters for both process interpretation and model-development realism (RQ1, RQ3).

I did *not* choose the other methods for pragmatic reasons, not because they were invalid:

- **M1 (schema validation)**: useful as a quick technical gate, but less informative once basic schema integrity is confirmed; M2 gave richer diagnostic value.
- **M3 (ICD-11 format accuracy)**: relevant, but high missingness in coding fields limited immediate explanatory value for the selected RQs.
- **M4 (actor/role representativity)**: partially overlaps M11; I prioritized M11 because it gives clearer workload concentration outputs.
- **M5 (modality case mix)**: useful for domain realism, but narrowed scope; less central than urgency/shift representativity for this report's RQ emphasis.
- **M7 (temporal sequence conformance)**: informative, but parts of the timestamp-credibility issue were already captured via M10 batch-logging sensitivity.
- **M9 (modality-specific semantic checks)**: clinically meaningful, but not as directly tied to cross-RQ synthesis as the selected set.

The key trade-off was breadth versus interpretability. Running five methods allowed full RQ coverage while still leaving space for deep interpretation and synthesis in Deliverable 6.

3. **Results interpretation. Provide detailed interpretation of results from each chosen method (key results, interpretation, expectation check, and primary facets).**

Answer

Key outputs are summarised in Appendix B, Table 9, with supporting plots in Figures 2–5.

Method 2 (Completeness). The strongest result was extensive missingness in context fields (for example `study_uid` 134/150, `modality` 101/150). This matched my expectation that administrative/process fields would be more complete than rich clinical context. It means sequence-based process analysis is feasible, but finer explanatory analysis is weakened. This primarily informs the **Data**, **Source**, and **System** facets.

Method 6 (Representativity). Results showed clear skew (15 stat/urgent vs 2 routine; Day 11, Night 6, Evening 0). I expected some imbalance, but not this magnitude. The implication is that estimates of “typical” flow are conditional on a narrow case mix. This directly affects generalisation for RQ1 and especially RQ3. Primary facets: **Data**, **Task**, **Human**.

Method 8 (Critical communication conformance). Using a 15-minute threshold, 16 cases passed and 1 failed (`CXR-004` missing communication event). I expected high compliance, and that was broadly confirmed, but the failed critical case is still clinically significant: rare events matter in patient-safety analysis. This result primarily informs **Task**, **System**, and **Human**.

Method 10 (Turnaround time with batch-logging detection). The key finding was high sensitivity of TAT conclusions to timestamp credibility: mean total TAT changed from 147.93 minutes (raw) to 198.50 minutes (clean), with 6/17 cases flagged for suspected batch logging (35.29%). I expected some effect, but the size of the difference was larger than expected. This indicates that process “speed” may be underestimated if logging behavior is not accounted for. Primary facets: **Data**, **System**, **Task**.

Method 11 (Workload distribution). Events were concentrated in a few roles (radiographer 64, system 31, senior 31, clinician 17, resident 7). This aligned with expectation that work is unevenly distributed, but it also highlighted that role concentration can shape both observed bottlenecks and training-data balance. Primary facets: **Human**, **Task**, **System**.

Overall, the methods worked as intended and were mutually informative. The main surprise across methods was how much interpretation changes once temporal credibility and representativity limits are made explicit.

4. **Your proposed DQ dimension. Define the dimension, include code, explain motivation with concrete examples, RQ impact, and facet mapping.**

Answer

I proposed the custom DQ dimension **operational authenticity**: whether logged traces represent real clinical operations rather than test/training/administrative artifacts.

What motivated this dimension? During manual exploration, one case (TEST-001) stood out with multiple contamination signals: non-standard case ID format, explicit training/test marker in notes (TRAINING CASE - DO NOT PROCESS), administrative actor involvement, and only a one-event trace. These patterns are not fully captured by the pre-implemented methods but can significantly bias downstream interpretation.

Custom assessment method code

Listing 1: Custom assessment method for operational authenticity

```
1 def my_custom_dq_method(df: pd.DataFrame) -> Dict[str, any]:
2     """
3     Assess operational authenticity (production-relevance validity).
4     """
5     rows = []
6
7     for case_id, case_df in df.groupby('case_id'):
8         reasons = []
9
10        if not re.match(r'^CXR-\d{3}$', str(case_id)):
11            reasons.append('Unexpected case_id format')
12
13        if case_df['notes'].astype(str).str.contains(
14            r'training|test|do not process', case=False, na=False
15        ).any():
16            reasons.append('Training/test markers in notes')
17
18        if case_df['actor'].eq('Admin User').any():
19            reasons.append('Administrative/test actor involvement')
20
21        if len(case_df) < 3:
22            reasons.append('Extremely short trace (low operational realism)')
23
24        score = len(reasons)
25        if score >= 3:
26            risk_level = 'high'
27        elif score >= 1:
28            risk_level = 'medium'
29        else:
30            risk_level = 'low'
31
32        rows.append({
33            'case_id': case_id,
34            'risk_level': risk_level,
```

```

35         'risk_score': score,
36         'reasons': '; '.join(reasons) if reasons else 'None',
37         'event_count': len(case_df),
38     })
39
40     case_flags = pd.DataFrame(rows).sort_values(
41         ['risk_score', 'case_id'], ascending=[False, True]
42     )
43     flagged = case_flags[case_flags['risk_level'].isin(['high', 'medium'])].copy()
44
45     summary_metrics = {
46         'total_cases': int(len(case_flags)),
47         'flagged_cases': int(len(flagged)),
48         'flagged_case_rate_pct': round(
49             100 * len(flagged) / len(case_flags), 2
50         ) if len(case_flags) > 0 else 0.0,
51         'high_risk_cases': int((case_flags['risk_level'] == 'high').sum()),
52     }
53
54     return {
55         'case_flags': case_flags,
56         'flagged_cases': flagged,
57         'summary_metrics': summary_metrics,
58     }

```

The result summary (Appendix B.2, Table 10) shows 1 flagged high-risk case out of 17 (5.88%). Even though this is a small proportion, it affects all RQs:

- **RQ1:** test traces can distort apparent workflow timing and throughput.
- **RQ2:** non-clinical traces can alter conformance denominators and produce misleading pass/fail rates.
- **RQ3:** contaminated training data can degrade model validity and external usefulness.

Facet mapping for this custom dimension is included in Appendix B.2, Table 11.

5. Per-Facet Data Quality Synthesis. For each facet, assess findings, main issues, and consequences for RQs. Then discuss the most challenging facet and alternative assessments.

Answer

This synthesis uses the selected-method outputs in Appendix B, Table 9, plus the custom-dimension outputs (Appendix B.2).

Data facet. Main issues were high attribute missingness (M2) and strong sensitivity to timestamp quality (M10). Consequence: RQ1 performance metrics and RQ3 feature quality are less robust than raw metrics suggest.

Source facet. The custom-dimension findings indicate provenance contamination risk (for example TEST-001), and M2 suggests that some source-dependent fields are inconsistently populated. Consequence: reduced confidence in transferability and training-data validity, especially for RQ3.

System facet. M8 and M10 show that system logging behavior shapes interpretation: one critical communication failure and substantial batch-logging signals. Consequence: RQ1 and RQ2 conclusions can be biased if system-generated timestamps are treated as perfect proxies for real-time work.

Task facet. M6 showed urgency/shift skew, and M8 depends on policy assumptions (for example 15-minute threshold). Consequence: RQ conclusions are condition-dependent; what looks compliant or efficient may partly reflect case-mix and threshold choices.

Human facet. M11 showed strong role concentration, and custom/M10 patterns suggest documentation behavior affects trace realism. Consequence: both workflow diagnosis (RQ1) and AI-development assumptions (RQ3) can be biased by who documents and when.

Which facet was most challenging? The **Human** facet was hardest to assess using available methods. Logs show recorded actions but rarely capture intent, negotiation, informal checks, or silent coordination. In practice, the **Source** facet was also difficult for similar reasons (provenance often requires context beyond event rows).

How else could facets be assessed?

- Add explicit provenance metadata (production/test flag, entry mode, user context).
- Triangulate with linked systems (RIS/PACS/message logs) for cross-source validation.
- Use targeted mini-interviews or debriefs for anomalous cases.
- Run periodic audits on documentation practice and timing behavior.

6. **Invisible work and system design implications. Use the Part 7 analysis and workflow diagram to address missing activities, reasons, system-design consequences, Part I parallel, and one concrete logging-system change with trade-offs.**

Answer

The invisible-work comparison table is provided in Appendix B.5, Table 13, based on the workflow diagram (Appendix B.4, Figure 6). Key examples include radiologist-to-

radiologist consultation, clarification calls to ordering clinicians, informal reprioritisation/handoffs, and cross-team patient-readiness coordination.

These activities are likely absent from the event log for combined **technical**, **social**, and **economic** reasons:

- Technical: much coordination happens by phone, hallway discussion, or across systems without a shared event schema.
- Social: informal coordination is situational and often intentionally lightweight.
- Economic/workload: full documentation of all micro-interactions adds time burden and may conflict with clinical throughput.

If a process model is built only from logged events, important distortions follow:

- **RQ1:** delays can be misattributed to “visible” stages while hidden coordination delays remain invisible.
- **RQ2:** safety-critical communication may appear more compliant than practice if verbal escalation pathways are not represented.
- **RQ3:** AI models trained on log-only traces may learn an oversimplified process that omits real coordination dynamics.

This parallels my Part I experience: interview-based models captured an orderly high-level routine, while observation captured practical deviations and coordination work. The event-log vs workflow-diagram gap is the same phenomenon at a different scale: *work-as-recorded* is not identical to *work-as-done*.

The broader implication is that data-driven design is necessary but not sufficient in health-care. Following Chapter 9’s formal/informal system distinction, formal logs capture what is codified, while critical informal coordination can remain outside system boundaries (Guide to Health Informatics 2026; Guide to Health Informatics 2026).

One concrete improvement is to add a lightweight `coordination_interaction` event type at selected high-impact points (for example escalation, reprioritisation, cross-team hand-off), with minimal required fields: timestamp, linked case ID, initiating role, receiving role, and reason code. The trade-off is extra documentation load and potential surveillance concerns. A practical mitigation is selective capture (not universal capture), role-based visibility, and periodic review to ensure data are used for workflow improvement rather than individual blame.

A Supporting Tables for Part I

This appendix contains supplementary Part I material referenced from Deliverables 1–4.

Table 2: Observation guide used during Deliverable 2

Focus	Observation prompt	Evidence target
Sequence and boundary	Which event marks start/end, and what is the observed order of actions?	Event IDs, timestamps, sequence notes
Tools and artefacts	Which tools/resources are used at each step?	Tool field, context notes
Interactions and coordination	Does any interaction with other people occur?	Actor field, coordination notes
Deviations from interview baseline	Which actions are skipped, added, reordered, or relabeled relative to A01-A09?	Related activity ID + delta notes
Adaptation and recovery	Which workaround or exception handling occurs under time/resource pressure?	Context notes for exception events

Source: Zajac (2026) and Guide to Health Informatics (2026a): interview + observation + process-tracing principles adapted to this routine study.

Note: Field notes were split into descriptive and reflexive notes; only descriptive notes were transcribed into the submitted CSV event log.

A.1 Core Deliverable Artefacts (Part I)

Table 3: Line-numbered self-interview transcript (conducted February 18, 2026)

Line	Transcript
L01	Interview question: Please describe your normal weekday morning routine from when you wake up until you leave home.
L02	Participant: My alarm goes off at 06:45, and I usually get up right away after stopping it.
L03	Participant: I first drink a glass of water, then I go to the bathroom.
L04	Participant: I normally take a quick shower, around ten to twelve minutes.
L05	Participant: After that I get dressed and prepare my bag for the day.
L06	Participant: I usually pack my laptop, charger, student ID card, wallet, and keys.
L07	Participant: Then I make coffee and toast.
L08	Participant: I normally sit down, eat breakfast, and check my calendar and transport timing on my phone.
L09	Participant: After breakfast I brush my teeth and do a final mirror check.
L10	Participant: I check the weather so I can choose the right jacket.
L11	Participant: Before leaving, I do a final essentials check: phone, wallet, keys, and card.
L12	Participant: I aim to leave at around 07:30 to catch transport to campus.
L13	Interview question: Which steps vary the most from day to day?
L14	Participant: Sometimes I snooze once if I am tired.
L15	Participant: Breakfast can change; if I am in a rush, I might just eat fruit instead of toast.
L16	Participant: Occasionally I spend a minute looking for keys if I did not put them in the same place.
L17	Interview question: Are other people typically involved in this routine?
L18	Participant: Mostly no, I do it alone, but once in a while I text my roommate if I need a shared item.
L19	Interview question: Which tools or systems are central in the routine?
L20	Participant: My phone alarm, calendar, weather app, the coffee machine, toaster, and my backpack.
L21	Interview question: How do you define start and end of this routine?
L22	Participant: It starts when the alarm sounds and ends when I close the apartment door behind me.

Table 4: Interview-based activity model (work-as-imagined / knowledge-as-stated)

ID	Activity label	Actor	Order	Expected time	Main tools/resources
A01	Stop alarm and orient to time	Participant	1	06:45-06:47	Smartphone alarm
A02	Drink water	Participant	2	06:47-06:50	Glass, tap
A03	Toilet and shower	Participant	3	06:50-07:02	Bathroom, shower
A04	Dress and pack bag	Participant	4	07:02-07:12	Wardrobe, backpack
A05	Prepare breakfast and coffee	Participant	5	07:12-07:20	Coffee machine, toaster
A06	Eat breakfast and review schedule	Participant	6	07:20-07:25	Phone calendar/transport apps
A07	Brush teeth and final grooming	Participant	7	07:25-07:28	Toothbrush, mirror
A08	Check weather and essentials	Participant	8	07:28-07:30	Weather app, keys, wallet, ID
A09	Leave apartment for commute	Participant	9	07:30	Front door, bag

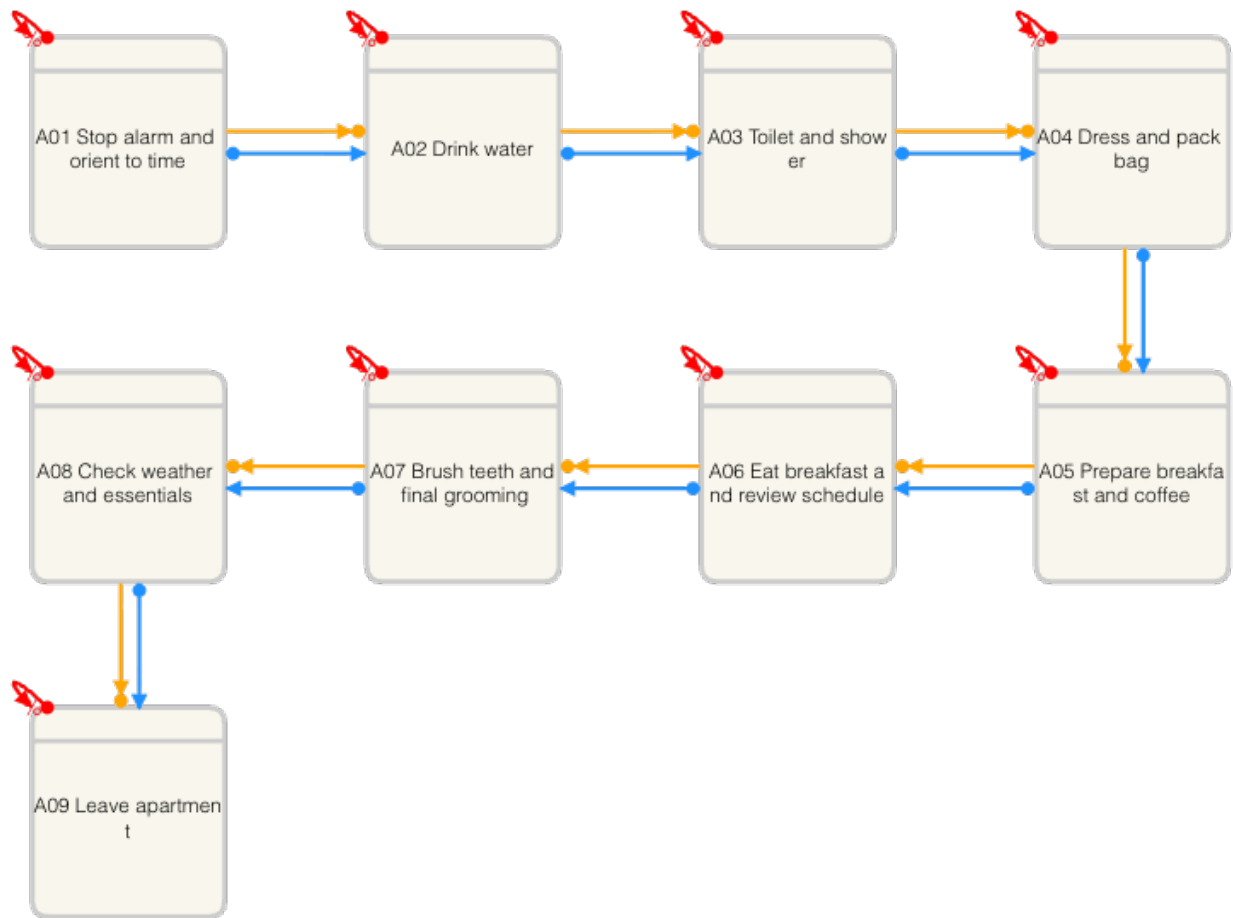


Figure 1: Interview-based DCR graph for the morning routine (A01-A09)

Source: Own DCR Graph export (February 2026), built from transcript evidence in Table 3 (lines L02-L12, L14-L18, L21-L22).

Note: Used as baseline work-as-imagined model for comparison with observed events in Deliverables 2 and 3.

Table 5: Observed event-log excerpt (February 20, 2026)

Event	Timestamp	Event label	Tool	Context note	Ref.
E01	2026-02-20 06:45	Alarm rings and snooze	Smartphone alarm	One snooze before wake-up.	A01
E03	2026-02-20 06:52	Reply to roommate message	Messaging app	Extra micro-coordination event.	NA
E05	2026-02-20 06:57	Toilet and shower	Bathroom	Hygiene block started later than interview expectation.	A03
E07	2026-02-20 07:12	Start coffee and toast	Coffee machine, toaster	Breakfast preparation begins.	A05
E08	2026-02-20 07:16	Switch breakfast to banana	Kitchen	Breakfast adapted due to no bread.	A05
E09	2026-02-20 07:18	Pack bag	Backpack	Packing occurs after breakfast prep starts.	A04
E10	2026-02-20 07:21	Find keys and borrow lock	Keys, messaging app	Workaround to solve missing item.	A08
E14	2026-02-20 07:30	Leave apartment	Front door	End of observed routine boundary.	A09

Table 6: Delta sheet between interview model and observed routine

Delta ID	Category	Interview ref.	Observed ref.	Difference description	Implication
D01	Missing events	A06	No direct event	No distinct seated breakfast-and-schedule-review event occurred.	Interview model overstates routine regularity.
D02	Extra events	None in baseline	E03 (06:52)	Short roommate-message coordination appears in observed routine only.	Reveals practical coordination load.
D03	Different order	A04 before A05	E07 then E09	Bag packing happened after breakfast preparation started.	Ordering constraints are weaker than stated.
D04	Timing distortions	A03 expected 06:50-07:02	E05 start 06:57	Hygiene block shifted later due to wake-up delays.	Time windows in interview are optimistic.
D05	Workarounds/coordination steps	A08 final check	E10 (07:21)	Missing lock triggered ad hoc borrowing via message.	Recovery actions are underrepresented in interview model.
D06	Relabeling/ambiguity	A05/A06	E08, E13	“Breakfast” changed from sit-down meal to banana plus coffee-to-go.	Label definitions need operational clarity.

B Supporting Tables and Code for Part II

This section includes only the relevant outputs from the completed Deliverable 5 notebook (`assignment.ipynb`): selected method evidence, key visualisations, custom DQ-dimension results, and invisible-work comparison material.

Table 7: Part II log profile from notebook Part 1 exploration

Metric	Result
Total events	150
Unique cases	17
Time range covered	2026-01-12 07:45 to 2026-01-13 08:46
Unique logged activities	12
Columns with strongest missingness	study_uid (134), series_count (134), instance_count (134), icd11_code (131), report_id (113), modality (101), notes (73)

Table 8: Selected methods run in notebook Part 4

Method	DQ dimension	Primary facets	RQ link	Selection rationale
M2	Completeness	Data, Source, System	RQ1–RQ3	Baseline quality gate before downstream interpretation.
M6	Representativity	Data, Task, Human	RQ1, RQ3	Checks urgency and temporal balance of case coverage.
M8	Timeliness conformance	Task, System, Human	RQ2	Tests safety-critical communication behavior directly.
M10	Timeliness + credibility	Data, System, Task	RQ1	Quantifies TAT and sensitivity to batch logging.
M11	Workload representativity	Human, Task, System	RQ1, RQ3	Reveals concentration across actors/roles.

Table 9: Key quantitative outputs from selected methods

Method	Key output	Relevance for Deliverable 6 reflection
M2	Context-field missingness is high (e.g., study_uid 134/150; modality 101/150).	Shows what can and cannot be trusted for workflow and AI-oriented interpretations.
M6	Urgency skew (15 stat/urgent, 2 routine) and shift imbalance (Day 11, Night 6, Evening 0).	Demonstrates representativity limits for RQ1 and especially RQ3 generalisation.
M8	Critical communication: 16 pass, 1 fail (CXR-004 missing communication event).	Direct patient-safety signal tied to RQ2 and conformance assumptions.
M10	Mean total TAT = 147.93 min raw vs 198.50 min clean; 6/17 cases flagged as batch-logged (35.29%).	Shows strong sensitivity of performance conclusions (RQ1) to timestamp credibility.
M11	Events per role: radiographer 64, system 31, senior 31, clinician 17, resident 7.	Quantifies workload concentration and role imbalance relevant to RQ1 and RQ3.

B.1 Notebook Visualisations (Relevant Outputs)

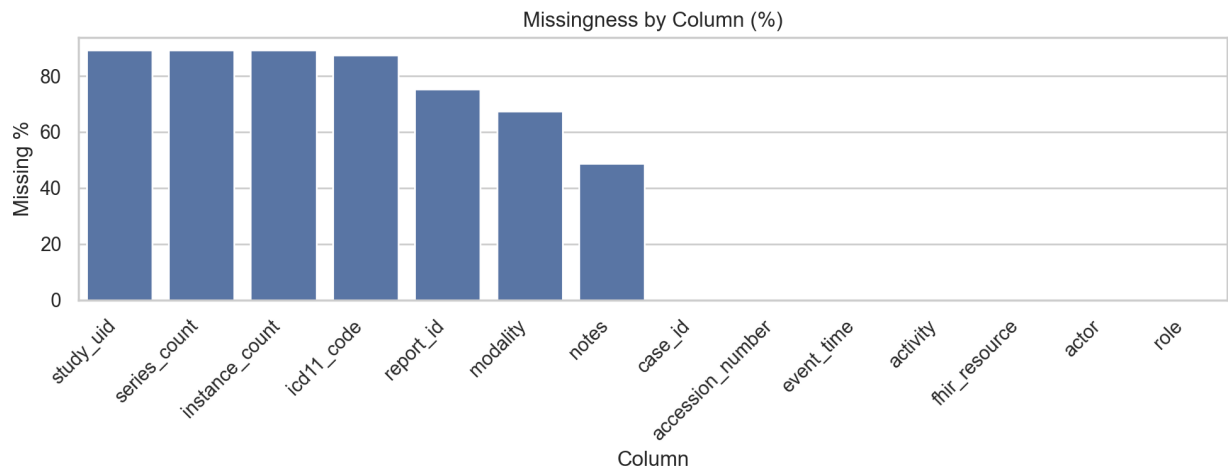


Figure 2: Method 2 output: missingness profile by column

Source: Generated from Deliverable 5 notebook output using *radiology_event_log.csv*.

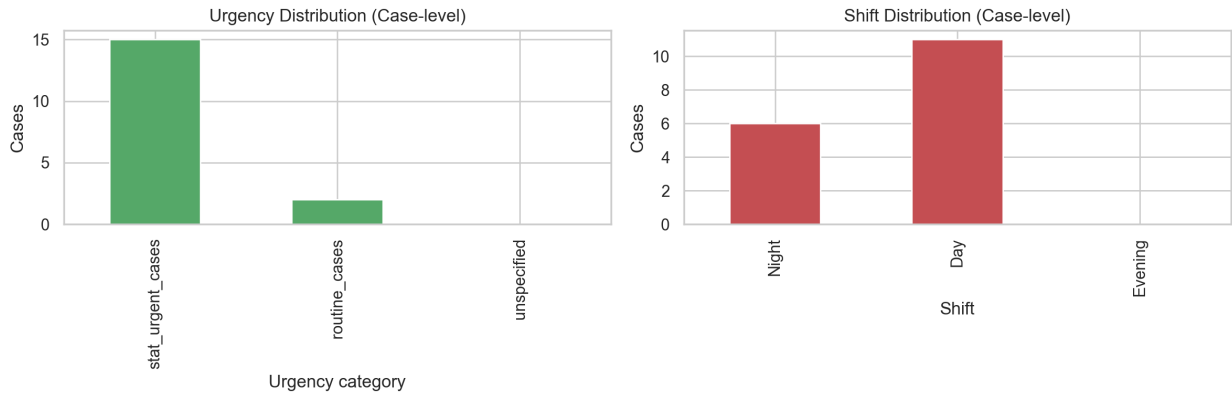


Figure 3: Method 6 output: urgency and shift representativity

Source: Generated from Deliverable 5 notebook output using *radiology_event_log.csv*.

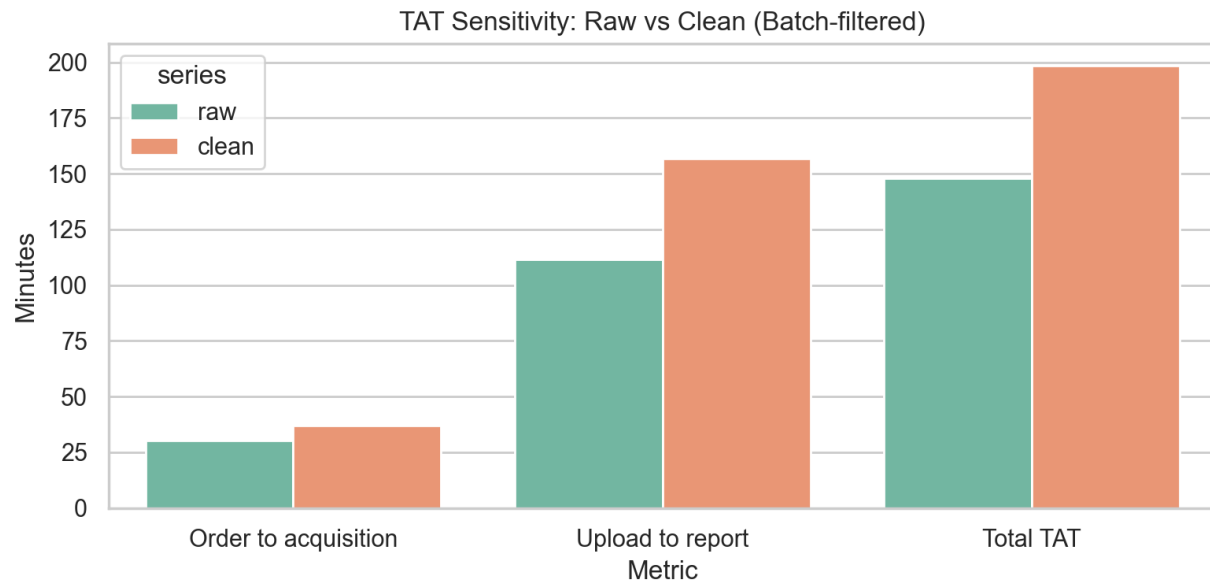


Figure 4: Method 10 output: turnaround-time sensitivity (raw vs batch-filtered)
Source: Generated from Deliverable 5 notebook output using *radiology_event_log.csv*.

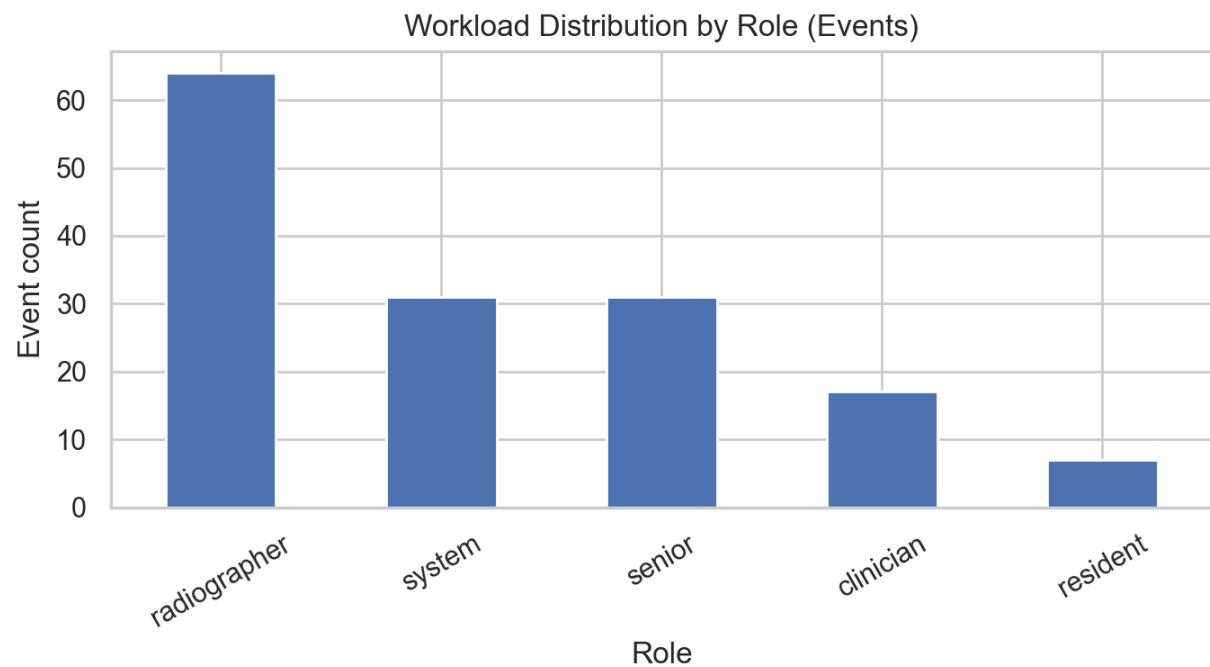


Figure 5: Method 11 output: workload distribution by role
Source: Generated from Deliverable 5 notebook output using *radiology_event_log.csv*.

B.2 Custom DQ Dimension (Notebook Part 5)

Table 10: Custom DQ dimension result summary: operational authenticity

Metric	Result
Total evaluated cases	17
Flagged cases (medium/high risk)	1
Flagged case rate	5.88%
High-risk cases	1
Flagged case details	TEST-001: unexpected case ID format; training/test marker in notes; administrative actor involvement; extremely short trace (1 event)

Table 11: Facet mapping for custom dimension: operational authenticity

Facet	Involvement	Justification
Data	++	Case identifiers and note text contain direct contamination signals (test/training traces).
Source	++	Source governance determines whether non-production traces enter the analysis dataset.
System	+	Logging architecture does not fully separate test and production provenance.
Task	+	KPI and triage conclusions become biased when non-clinical traces are included.
Human	+	Human setup/maintenance actions can introduce non-operational events.

B.3 Custom DQ Assessment Method Code

Listing 2: Custom DQ assessment method from notebook Part 5 (operational authenticity)

```
1 def my_custom_dq_method(df: pd.DataFrame) -> Dict[str, any]:
2     """
3     Assess operational authenticity (production-relevance validity).
4
5     Goal:
6     Identify case traces that are likely test/training/non-production records
7     or otherwise not representative of real clinical workflow data.
8     """
9     rows = []
10
11     for case_id, case_df in df.groupby('case_id'):
12         reasons = []
13
14         if not re.match(r'^CXR-\d{3}$', str(case_id)):
```

```

15         reasons.append('Unexpected case_id format')
16
17     if case_df['notes'].astype(str).str.contains(
18         r'training|test|do not process', case=False, na=False
19     ).any():
20         reasons.append('Training/test markers in notes')
21
22     if case_df['actor'].eq('Admin User').any():
23         reasons.append('Administrative/test actor involvement')
24
25     if len(case_df) < 3:
26         reasons.append('Extremely short trace (low operational realism)')
27
28     score = len(reasons)
29     if score >= 3:
30         risk_level = 'high'
31     elif score >= 1:
32         risk_level = 'medium'
33     else:
34         risk_level = 'low'
35
36     rows.append({
37         'case_id': case_id,
38         'risk_level': risk_level,
39         'risk_score': score,
40         'reasons': '; '.join(reasons) if reasons else 'None',
41         'event_count': len(case_df),
42     })
43
44     case_flags = pd.DataFrame(rows).sort_values(
45         ['risk_score', 'case_id'], ascending=[False, True]
46     )
47     flagged = case_flags[case_flags['risk_level'].isin(['high', 'medium'])].copy()
48
49     summary_metrics = {
50         'total_cases': int(len(case_flags)),
51         'flagged_cases': int(len(flagged)),
52         'flagged_case_rate_pct': round(
53             100 * len(flagged) / len(case_flags), 2
54         ) if len(case_flags) > 0 else 0.0,
55         'high_risk_cases': int((case_flags['risk_level'] == 'high').sum()),
56     }
57
58     return {
59         'case_flags': case_flags,

```

```

60     'flagged_cases': flagged,
61     'summary_metrics': summary_metrics,
62 }

```

B.4 Workflow Diagram from Assignment Brief

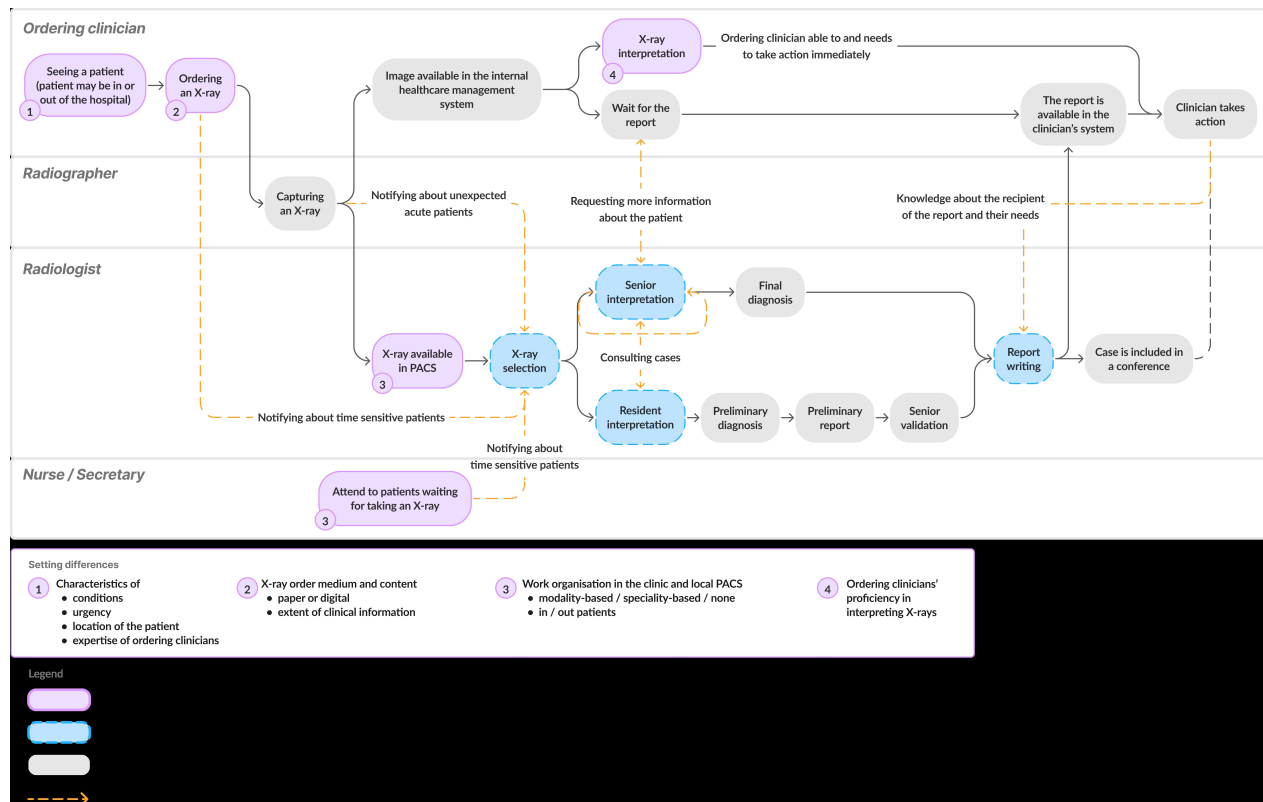


Figure 6: Assignment Figure 1 from the Assignment 2 PDF: observed radiology workflow with formal and informal coordination paths

Source: Provided in Assignment 2 brief as Figure 1 (Figur1-assignment2.jpg).

B.5 Invisible Work Material (Notebook Part 7)

Table 12: Activities logged in the event log (12 unique activities)

Logged activity set A	Logged activity set B
Case selected	Order placed
Critical communication	Patient selected from worklist
Image acquisition start	Preliminary report created
Image acquisition end	Report distributed
Images uploaded to PACS	Report finalised
Order accepted	Report validated and approved

Table 13: Invisible-work activities from workflow diagram not represented explicitly in log

Activity from diagram	Why not logged?	Facet	Consequence	Capture option
Radiologist-to-radiologist consultation	Mostly verbal/ad hoc interaction outside structured capture	Human, Task	Collaborative interpretation work is under-represented	Optional lightweight consult event
Clarification call to ordering clinician	Usually done by phone outside RIS/PACS event schema	System, Source	Communication-delay attribution becomes incomplete	Linked communication event type
Informal reprioritisation and handoff	Queue coordination is locally negotiated rather than formalised	Task, Human	Bottleneck and fairness analyses miss practical prioritisation logic	Priority-change event with rationale
Cross-team coordination on patient readiness	Interdepartmental coordination spans systems and roles	Source, System, Task	Waiting-time drivers become invisible in case timeline analysis	Interdepartment status event

B.6 Textbook Figures Referenced in the Report

The figures below are included to support direct references in Part I and Part II. To separate them from assignment-produced figures, each caption explicitly preserves the textbook figure number.

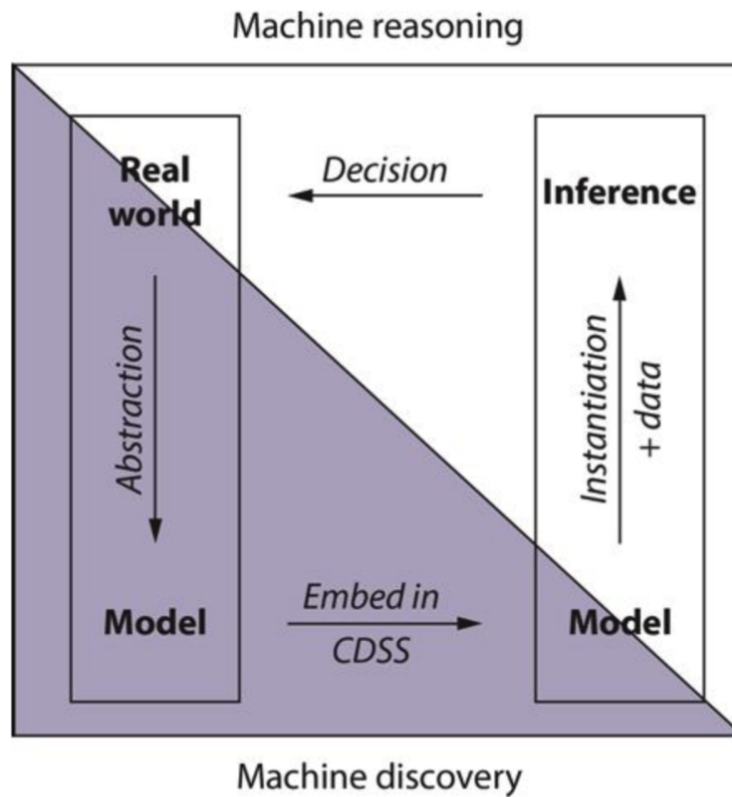


Figure 27.1 Computer reasoning depends first on acquiring models that record our understanding of how the world works and then applying those models in a decision support system to arrive at conclusions shaped by the data specific to a given decision task.

Figure 7: Textbook Figure 27.1 from *Guide to Health Informatics*: computer reasoning cycle from model construction through use

Source: *Guide to Health Informatics* (2026a). Reproduced from course compendium reading.

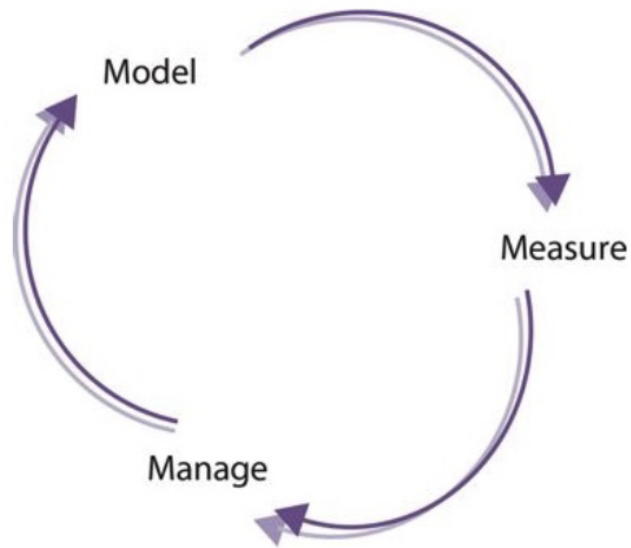


Figure 9.1 The model-measure-manage cycle. When the outcome of a management decision is fed back into another cycle, a feedback control loop is formed.

Figure 8: Textbook Figure 9.1 from *Guide to Health Informatics*: model-measure-manage cycle
Source: *Guide to Health Informatics* (2026b). Reproduced from course compendium reading.

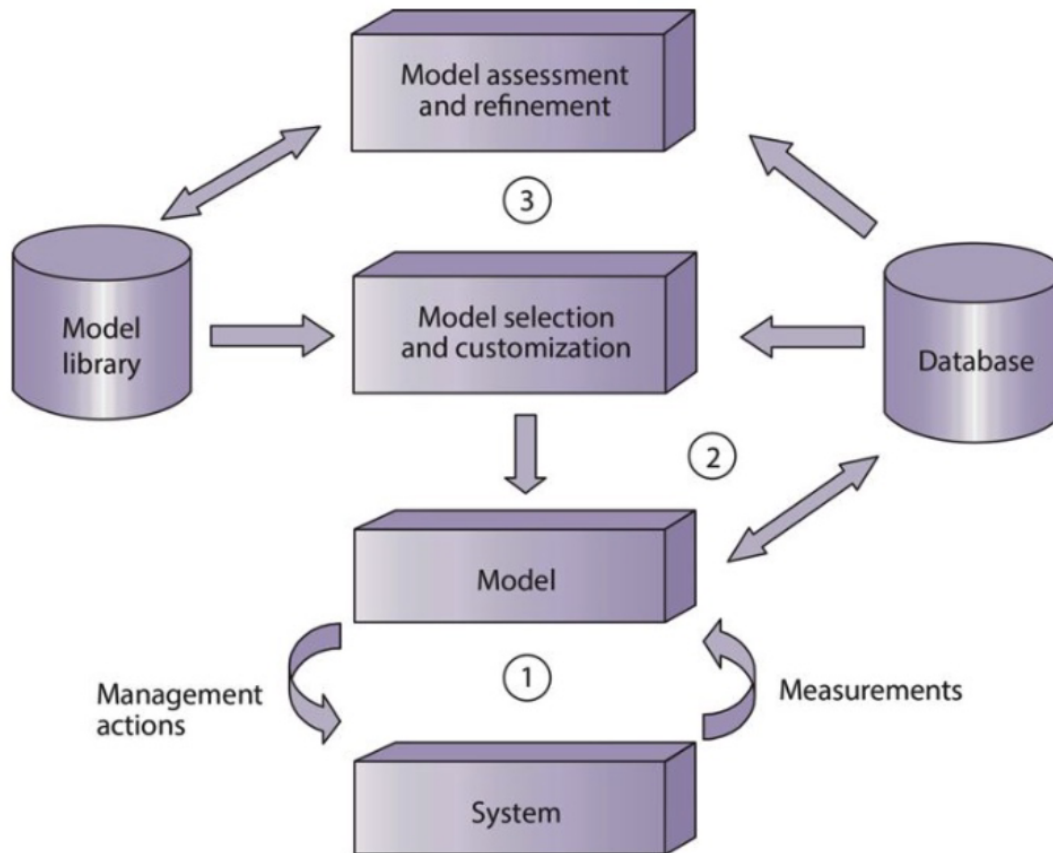


Figure 9.2 The three-loop model. Three separate information cycles interact within the health system. The first and second cycles select and apply models to the management of specific systems. The third information cycle tries to assess the effectiveness of these decisions and improve the models accordingly.

Figure 9: Textbook Figure 9.2 from *Guide to Health Informatics*: three-loop model
 Source: *Guide to Health Informatics* (2026b). Reproduced from course compendium reading.

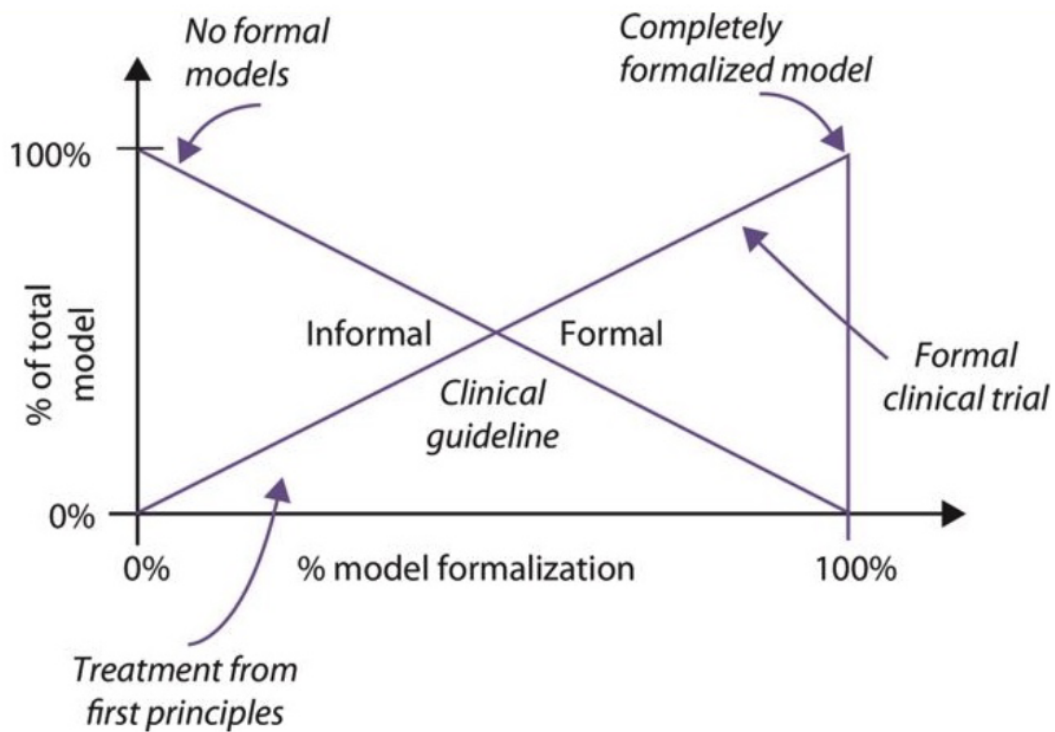


Figure 9.3 There is a continuum of possible formalization of information systems. Formal and informal models combine together to cover the total modelling needs of a given problem. The degree of formalization depends on the type of problem being addressed.

Figure 10: Textbook Figure 9.3 from *Guide to Health Informatics*: formalization continuum
 Source: *Guide to Health Informatics* (2026b). Reproduced from course compendium reading.

References

- Guide to Health Informatics (Course Compendium) (2026a). *Guide to Health Informatics: Selected Chapter 27*. PDF. Course reading PDF provided in NDAB20010U.
- (2026b). *Guide to Health Informatics: Selected Chapters 9 and 27*. PDF. Course reading PDF provided in NDAB20010U.
- Zajac, H. D. (Feb. 2026). *Sundhedsdata og interoperabilitet: Methods of knowing + Health-care data*. PDF. Course slides, NDAB20010U Health Data and Interoperability, University of Copenhagen.